

PROJECT COMPLETION DOCUMENT

HOTEL RESERVATION SYSTEM

A Java-Based Console Application Demonstrating Object-Oriented Programming, Collections, File Handling, and Exception Handling

Field	Details
Project Title	Hotel Reservation System
Technology	Java (Console Application)
Development Approach	Object-Oriented Programming (OOP)
Status	Completed

1. Introduction

The Hotel Reservation System is a Java-based console application developed using Object-Oriented Programming concepts. The project automates hotel room booking, customer registration, room availability checking, bill generation, and reservation management.

The system uses Java Collections for storing data, File Handling for saving reservations, and Exception Handling to prevent runtime errors. This document certifies the successful completion of the project and summarizes its design, implementation, and outcomes.

2. Project Objectives

- Automate hotel room booking
- Store customer information
- Display available rooms
- Generate customer bills
- Save reservation details into a file
- Demonstrate OOP concepts
- Demonstrate the Collections Framework
- Demonstrate File Handling
- Demonstrate Exception Handling

3. Software & Hardware Requirements

3.1 Hardware Requirements

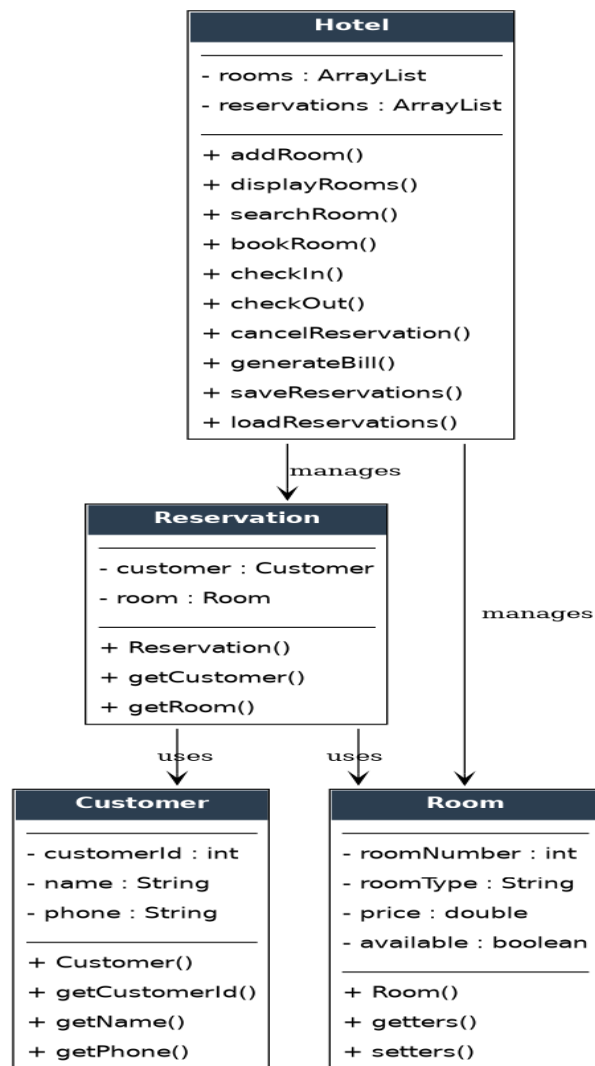
- Processor: Intel i3 or above
- RAM: 4 GB
- Hard Disk: 100 MB free space

3.2 Software Requirements

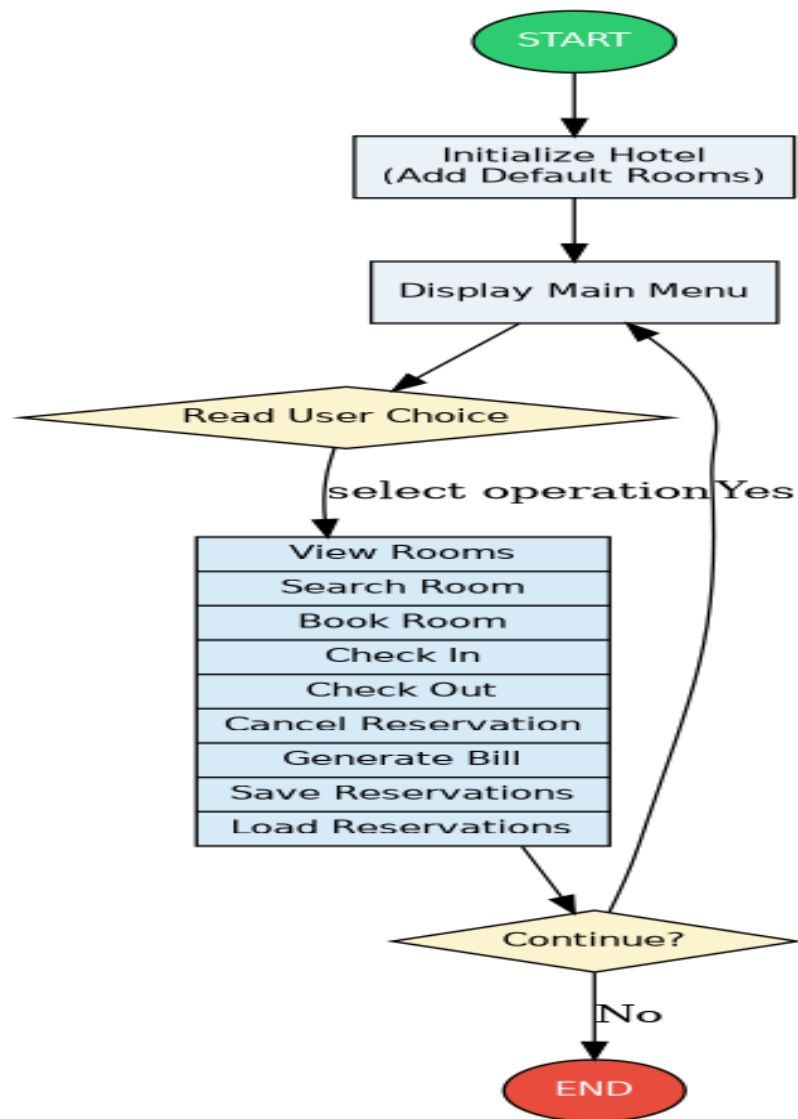
- Java JDK 8 or above
- Eclipse IDE
- Windows / Linux

4. Project Modules

Module	Functions & Class Used
Room Management	Add Room, Display Rooms, Search Room — Class: Room.java
Customer Registration	Register Customer, Store Customer Information — Class: Customer.java
Reservation	Book Room, Cancel Reservation — Class: Reservation.java
Hotel Management	Check-In, Check-Out, Bill Generation, Save/Load Reservations — Class: Hotel.java
User Interface	Menu, User Input, Method Calls — Class: Main.java



UML Class Diagram



Process Flowchart

5. Object-Oriented Programming Concepts Used

5.1 Encapsulation

Private variables are accessed through getter and setter methods.

```
private int roomNumber;
```

5.2 Classes

- Room
- Customer
- Reservation
- Hotel
- Main

5.3 Objects

```
Customer c = new Customer(1, "John", "9876543210");
```

5.4 Constructors

```
Room(int roomNumber, String roomType, double price)
```

6. Collections Framework

The project uses ArrayList to manage dynamic collections of rooms and reservations.

```
ArrayList<Room> rooms = new ArrayList<>();
```

Advantages:

- Dynamic size
- Easy insertion
- Easy deletion

7. Exception Handling

Exception handling is used to prevent the program from crashing due to invalid input.

```
try {    int choice = sc.nextInt(); } catch (InputMismatchException e) {  
System.out.println("Invalid Input"); }
```

8. File Handling

Classes used: FileWriter, FileReader, BufferedReader.

```
FileWriter writer = new FileWriter("reservations.txt");
```

Purpose: Stores reservation information permanently so it persists across program runs.

9. Algorithm

1. Start the program
2. Create the Hotel object
3. Add rooms
4. Display the menu
5. Read the user's choice
6. Perform the requested operation
7. Repeat until Exit is selected
8. Stop

10. Sample Output

10.1 Main Menu

```
HOTEL RESERVATION SYSTEM 1. View Rooms 2. Search Room 3. Book Room 4. Check In 5.  
Check Out 6. Cancel Reservation 7. Generate Bill 8. Save Reservations 9. Load  
Reservations 10. Exit
```

10.2 Booking Example

Customer ID : 1 Name : David Phone : 9876543210 Room Number : 102
Room Booked Successfully.

10.3 Bill Generation

Room Number : 102 Days Stayed : 5 Total Bill = \$400

10.4 Save Reservation

Saved Successfully.

11. Advantages

- User-friendly
- Easy to understand
- Fast room booking
- Prevents duplicate booking
- Saves reservation records
- Demonstrates core Java concepts

12. Limitations

- Console-based interface
- No database connectivity
- No login/authentication
- Reservation data is stored in a text file only

13. Future Enhancements

- MySQL database integration
- Java Swing/JavaFX GUI
- Online payment system
- Customer login
- Admin dashboard
- Room images
- Date-based reservations
- Email/SMS notifications

14. Conclusion

The Hotel Reservation System successfully automates hotel booking operations using Java. The project demonstrates Object-Oriented Programming, Collections, Exception Handling, and File Handling in a practical application. It is simple, reliable, easy to maintain, and suitable as a Java mini-project for academic purposes.

15. Project Completion Statement

This is to certify that the Hotel Reservation System project has been designed, developed, and tested successfully, meeting all the objectives outlined in Section 2 of this document. All modules — Room

Management, Customer Registration, Reservation, Hotel Management, and the User Interface — have been implemented and verified to function as intended.